3-12 Cardiac ischaemia v.1

If the patient is unconscious, signs of cardiac ischaemia primarily include:

- ST elevation or depression
- T wave flattening or inversion
- Arrhythmias, particularly ventricular
- Other haemodynamic abnormalities (hypo- or hypertension, tachy- or bradycardia)
- New or evolving regional wall motion abnormalities if echocardiography is used

If the patient is conscious, symptoms may include chest pain, breathlessness, dizziness, nausea and vomiting.

Have a high index of suspicion in patients with a pre-existing history or risk factors for cardiac ischaemia

START

1. Call for cardiac arrest trolley and 12-lead ECG machine.
2. Ensure adequate oxygenation and anaesthesia/analgesia.
3. Treat haemodynamic instability (Box A).
4. Apply CM5 continuous ECG monitoring (Box B). Obtain a 12-lead ECG as soon as possible.
5. If ischaemia does not resolve:
   - Call for help. Inform theatre team of problem. Stop or rapidly complete the surgery.
   - Start glyceryl trinitrate (GTN) (Box C).
   - EXTREME CAUTION with GTN if the patient is hypotensive.
6. Consider invasive arterial blood pressure monitoring.
7. Treat electrolyte abnormalities particularly potassium, magnesium and calcium.
8. Treat anaemia aiming for haematocrit >30%.
   - CAUTION – beware volume overload especially in heart failure.
9. If persistent ST elevation is present, consider need for anticoagulation, anti-platelet therapy and revascularisation in consultation with cardiology and surgical teams.

Box A: HAEMODYNAMIC INSTABILITY

- Cardiac arrest → **2-1**
- Hypotension → **2-4**
- Hypertension → **2-5**
- Bradycardia → **2-6**
- Tachycardia → **2-7**

Box B: CM5 ECG CONFIGURATION

- Right arm (red) lead over upper right sternum.
- Left arm (yellow) lead 5th intercostal space under left nipple.
- Indifferent (green or black) lead on left shoulder.

Box C: GLYCERYL TRINITRATE (GTN) DOSE

- Consider sublingual administration.
- i.v.: 1% (1 $mg.ml^{-1}$) solution – start at $0.1ml.kg^{-1}.hr^{-1}$, titrate against response.
- NOT RECOMMENDED IN CHILDREN.

Box D: AFTER THE EVENT

Admit to critical care environment and consult cardiology
Maintain head up position if practicable
Obtain serial 12-lead ECGs and cardiac enzymes

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